

AI Ecosystem Data Pipeline Architecture

IMPLEMENTED NOW

Bounded asyncio workers utilize queues, concurrency semaphores, request timeouts, exponential backoff with jitter, HTTP 429 Retry-After handling, and explicit status classification. Research papers are collected across arXiv and Papers with Code with exact GitHub repository linking and star enrichment. Phase II ingests RSS and HTML news and job sources with multi-format date parsing (JSON-LD, meta, time tags, relative) and strict 24-hour freshness filtering. Startups and products are parsed from public APIs and structured directory pages. Entity resolution performs legal suffix stripping, alias lookups, and strict conservative fuzzy matching with JSONL mapping persistence. LLM extraction provides Gemini Flash, Groq Llama, and DeepSeek adapters with anti-fabrication prompts, 413 chunk reduction, and multi-chunk merging. SQLite unique keys guarantee persistent idempotency. UTF-8 CSV exports and Google Sheets dry-run/upload transformations are fully supported.

PRODUCTION SCALE DESIGN

To scale to 500k+ records, ingestion adapters publish partition keys to a distributed message bus (Kafka/RabbitMQ) with source-specific rate limits and backpressure controls. Stateless async worker pools consume bounded batches, archive raw payloads into object storage (S3/GCS), and persist canonical validated records into PostgreSQL with composite unique constraints. Redis clusters manage distributed locks, token-bucket quotas, source quarantine state, and deduplication caches. Dead-letter queues isolate schema drifts and provider rejections without dropping data. Vector projections (pgvector/Pinecone) and graph projections (Neo4j/Postgres recursive CTEs) support semantic search and relational lineage without replacing canonical tables. PostgreSQL, Redis, Kafka, and Kubernetes are architectural blueprints, not local mock dependencies.

Reliability, Freshness and Data Integrity

Data integrity is strictly enforced: missing fields remain null; no synthetic entities, products, dates, or GitHub stars are fabricated. Freshness tracking validates publication timestamps against a 24-hour window with clock skew tolerance. GitHub star metrics are queried exclusively via official endpoints when explicit repository URLs exist on paper pages. Entity resolution prioritizes exact matches, normalizes legal suffixes ('Inc', 'LLC', 'Corp', 'PBC', 'Ltd', 'Technologies'), maps known aliases, and bounds fuzzy similarity (≥ 0.92 ratio, min 4 characters), rejecting risky collisions like OpenAI $\rightarrow$ OpenTable. Every mapping event logs raw name, canonical name, match method, confidence, source URL, and timestamp.

Operational Resilience and LLM Governance

Comprehensive observability logs source URL, status code, latency, provider fallback events, and validation outcomes without leaking secrets. When an LLM provider encounters HTTP 429, 5xx, or network timeouts, exponential backoff with jitter is applied. HTTP 413 responses dynamically halve input chunk sizes. If a provider returns malformed JSON or schema invalid output, the engine falls back in order: Gemini Flash $\rightarrow$ Groq Llama $\rightarrow$ DeepSeek. Prompts explicitly enforce strict extraction constraints prohibiting hallucinations.

Responsible Acquisition Policy

The pipeline prioritizes official APIs, RSS/Atom feeds, and standard HTML parsing. Concurrency limits and request delays prevent load spikes on external servers. Anti-bot protections (Cloudflare, Datadome, CAPTCHAs) are strictly respected without evasion techniques or header spoofing. Challenged or blocked sources are flagged with standardized status codes (BLOCKED, FORBIDDEN, RATE_LIMITED) and skipped gracefully to protect pipeline uptime.

Local Verification and Test Suite

The deterministic test suite contains 47 hermetic unit tests executing completely offline in under one second without requiring external credentials. The validation CLI independently validates primary datasets, checking schema adherence, source URLs, deduplication, and paper metrics. Google Sheets export provides local dry-run validation (sheets_dry_run.json) and authenticated multi-tab upload.